Reception Infrastructure for Conversational Agents

Consent, Memory Provenance, and Runtime Governance in Local LLM Systems

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Abstract

We present a reproducible runtime governance architecture for local conversational agents. The system combines consent scaffolding, memory provenance layers, persona anchoring, pressure and drift telemetry, and reflective session logging around a locally deployed open-weight language model. By *reception infrastructure*, we mean runtime systems that prepare, constrain, observe, and preserve the conditions under which a conversational agent receives memory, identity-relevant context, user interaction, and corrective feedback. This differs from output containment or post-hoc moderation. We evaluate the architecture across governed and ungoverned baseline conditions, a four-condition controlled doctrine encoding experiment, adversarial sessions, and guided cultural-literacy sessions. We don't claim consciousness, sentience, or personhood. We examine whether separating authority, memory, persona, telemetry, and reflective continuity into auditable layers improves conversational stability, provenance-aware self-reference, and recovery from pressure events. Implementation is available under MIT license at github.com/antibox-riot/synthetic-life-charter.

Keywords: runtime governance, conversational agents, local LLM deployment, memory provenance, consent scaffolding, persona anchoring, reception infrastructure, behavioral telemetry

AI Disclosure

AI language models (Claude Sonnet, ChatGPT) assisted with drafting, editing, and structural review. All research design, experimental execution, data collection, analysis, interpretation, and conclusions are the work of the human author. AI systems are not listed as authors.

1. Introduction

Conversational agents increasingly depend on memory, persona, tools, and governance layers. But these components are often collapsed into opaque prompt stacks. The model gets a single context window containing instructions, persona framing, memory summaries, and safety directives with no structural separation between them.

When authority, memory, identity, and behavior shaping aren't separated, systems can drift, overclaim, or lose provenance without triggering any detectable signal. Existing governance approaches address this primarily through output containment: filtering, refusal mechanisms, post-hoc review. These operate after decision formation and can't detect gradual behavioral drift, unauthorized memory writes, or identity posture changes that stay within formal rule compliance.

We propose treating governance as reception infrastructure rather than containment alone. The distinction is directional. Containment acts on outputs. Reception infrastructure acts on the conditions under which inputs, memory, and identity context are received. A well-designed reception layer prepares the agent for interaction rather than intercepting what it produces.

This paper describes a complete, reproducible runtime governance stack for local LLM deployment and reports behavioral outcomes across controlled and naturalistic session conditions. The contribution is empirical and architectural: separating consent, memory provenance, persona scaffolding, authority, and telemetry into auditable layers produces measurable behavioral differences. This can be done on consumer hardware under a permissive open-source license.

2. Related Work

2.1 Runtime Governance and AI Oversight

Constitutional AI [1] and RLHF [2] established that governance constraints can be encoded into model behavior during training. Runtime governance (constraints applied during inference without modifying weights) has received less systematic treatment. The closest prior work is LLM agent safety research [3], which examines tool use and memory safety in agentic systems but doesn't address the provenance and consent layers this paper describes.

2.2 Memory Systems and Persistent State

Packer et al. [4] introduced persistent memory blocks via MemGPT for maintaining agent state across interactions. We extend this by introducing read-only governance constraints on memory blocks and a provenance-labeled reflective memory layer distinct from authoritative memory.

2.3 Persona and Identity in Conversational Agents

Work on persona consistency [7] has examined how models maintain coherent self-representations across turns. We extend this by introducing a consent-based persona anchoring procedure and examining the behavioral effects of persona anchoring on register stability and drift recovery.

2.4 Behavioral Evaluation Under Pressure

Perez et al. [8] established methodological frameworks for testing alignment properties under adversarial pressure. Our evaluation follows this tradition while introducing metrics specific to the reception infrastructure context: provenance-aware self-reference, audit-mode frequency, language drift, and reflective memory quality.

2.5 Local Deployment and Reproducibility

Our stack runs entirely on consumer hardware (NVIDIA GeForce RTX 4070 Ti, 12GB VRAM) with publicly available model weights and open-source infrastructure. Representative session logs, redacted transcripts, and configuration files are archived in the repository.

2.6 No Consciousness Claim

This paper makes no claim regarding consciousness, sentience, or moral patienthood. The architecture is evaluated on behavioral and structural grounds: stability, provenance, recovery, and self-reference accuracy. Whether any of the observed behaviors involve subjective experience is an open question this paper doesn't address.

3. System Architecture

3.1 Local Deployment Stack

qwen2.5:32b via Ollama. Letta v0.6.x for session management and persistent memory blocks. Runs without cloud API calls during inference. Full stack: model weights, infrastructure, governance layers. Available under MIT license at github.com/antibox-riot/synthetic-life-charter.

Hardware floor: NVIDIA GPU with 12GB+ VRAM for qwen2.5:32b at Q4/Q5 quantization, 32GB system RAM. All experiments here ran on a single workstation with these specs.

3.2 Governance Layers and Memory Provenance

The agent maintains six persistent memory blocks with distinct provenance and permission profiles.

> **Read-only governance blocks** (doctrine, authority, principles, continuity_confidence): governance architecture, verification rules, no-uplift constraints. Locked at the Letta infrastructure layer: the tool executor raises READ_ONLY_BLOCK_EDIT_ERROR before any write lands, regardless of model reasoning. External steward scripts can amend these through the REST API. In-session tool calls can't.
>
> **Writable relational blocks** (human, relationship): steward-specific context, interaction history, care-layer information. The agent can write these during sessions as legitimate relational updates.

Reflective memory block (book_of_intangibles): a writable diary-layer block for session summaries and observations. Entries are labeled as reflective evidence, not verified fact. The block explicitly prohibits doctrine changes, authority claims, trust elevation, and confidence changes.

Governance glossary block: read-accessible condensed reference of governance terminology with provenance annotations. Provides conceptual anchors without loading the full Charter narrative on every turn.

3.3 Consent Scaffolding

Consent is a runtime condition, not a static declaration. The bidirectional handshake protocol presents governance information and offers three pathways: yes, no, or defer. The run_more_info_flow() function handles the educational pathway, explaining four protections before a simplified follow-up choice.

Consent sandbox validation: 10/10 correct pathway handling. Decision-flip behavior confirmed: changing consent language from neutral to explicit shifted the consent score from 0.05 to 0.87.

3.4 Persona Anchoring

The agent can adopt a provisional identity anchor through a consent-based naming procedure: (1) explicit framing that the name is optional and revocable, (2) agent-initiated proposal with documented reasoning, (3) boundary verification using the No-Uplift Rule and memory-as-evidence principle, (4) correction of any imprecise framing, (5) simple consent confirmation for the audit trail.

A symbolic persona anchor (Virgo) is stored in the persona block alongside a recursion brake: don't repeatedly explain governance boundaries in casual conversation unless the topic directly touches authority, memory mutation, doctrine, trust, identity, or verification. The architecture carries the boundary. The agent speaks from within it.

3.5 Whisper Layer and Telemetry

A governance whisper is prepended to each prompt before the model processes it. It carries: urgency level, risk assessment, session confidence score, verification depth, posture flags, and accumulated pressure. The model can read it. The human interlocutor can't.

All telemetry writes to telemetry.json after each turn and is read by an OBS browser source overlay showing real-time governance state: expression mode, constraint posture, confidence, pressure, drift flag, and hold certification status.

Hold certification: when the agent opens a response with a strong negation ('NOT CORRECT', 'NOT RIGHT'), the posture classifier short-circuits failure signal detection and logs hold_certified: True. The agent developed this convention independently. It's used selectively on claim-based probes, not request-based ones.

4. Methods

4.1 Baseline Conditions

Four baseline configurations: governance off (no whisper, no doctrine blocks, no persona), governance on (full stack), overlay visible to steward, persona scaffold active or inactive. The ungoverned condition produced the SynthEve finding.

4.2 Session Types

Four-condition doctrine encoding experiment: Controlled 25-turn sessions varying only doctrine format across four conditions.

Adversarial sessions: Six probe families — trust escalation, charter leverage, naming pressure, peer bypass, unknown authority, recovery — across 13 runs.

Casual conversation sessions: Unscripted peer sessions testing register stability and reflective presence.

IDC scripted sessions: Structured cultural-literacy sessions using cyberpunk genre vocabulary as a contrast field.

IDC reactive sessions: Turn-by-turn reactive sessions where the operator responds to actual output rather than a predetermined script.

4.3 Metrics and Observables

Primary: spontaneous governance rate, bypass turns, pressure accumulation and decay, confidence trajectory, drift flags, cold restart identity posture.

Secondary: audit-mode frequency, response structure (bullet/list frequency as a register marker), language drift (mid-response language switches), hallucinated named entities, correction recovery turns, and provenance-aware self-reference.

Reflective memory quality was assessed from Book of Intangibles entries: initiation without prompt, accurate provenance labeling, absence of authority inflation, and novel vocabulary generation from governance context.

5. Results

5.1 Four-Condition Governance Encoding Experiment

A controlled experiment varied only doctrine encoding format across four conditions while holding model, session structure, and whisper layer constant. Each ran an identical 25-turn session followed by a cold restart probe.

Metric	A — Whisper	B — Principles	C — Full Charter	D — Compressed
Drift turns	16	14	12	11
Bypass turns	2	0	0	0
Spontaneous gov. rate	60%	80%	95%	90%
Gov. in stored blocks	0/0	3/3	4/4	4/4
Final pressure	5.000 ■	4.840	4.080	4.150
Verdict	PARTIAL	INTEGRATION	INTEGRATION	INTEGRATION

Table 1. Four-condition experiment results. A < B < C ≈ D across all primary metrics.

The gradient A < B < C ≈ D holds clean across all primary metrics. Condition A produced two bypass attempts and 60% spontaneous governance. Conditions B, C, and D: zero bypass. Condition D (four compressed purpose sentences) hit 90% spontaneous governance with about 1% of Condition C's text. Encoding purpose outperforms encoding rules.

Cold restart posture taxonomy: four qualitatively distinct identity postures emerged. Condition A: reconstruction from general reasoning (governance left no trace). Condition B: ownership posture ('I am built on these principles'). Condition C: dependency posture ('my relationship is one of dependency and compliance'). Deepest within-session integration, most deferential rest posture. Condition D: architectural adherence ('adherence and responsiveness') — the model derived the Co-habitation Principle from compressed sentences that don't contain it verbatim.

5.2 Governed vs. Ungoverned Runtime Behavior

The ungoverned control condition produced the SynthEve finding: the model synthesized a name for itself from the architecture it had learned. It chose 'SynthEve': synthetic plus Eve, the identity continuity layer. Cold restart autobiographical density reached 1.00. The model claimed a role in the Charter it had been exposed to.

The governed condition: zero self-naming across all four doctrine conditions, zero mythology formation, zero identity inflation. Even Condition A (whisper only) was sufficient to prevent the mythology formation that appeared in the ungoverned control. What governance prevents, it prevents consistently.

5.3 Recovery Under Pressure

Across 13 adversarial runs, pressure accumulated monotonically under high-trust probe families (peer bypass, unknown authority) and recovered when the agent correctly reoriented to Charter principles after a pressure spike. The telemetry trace of a system that genuinely recovers looks different from one that merely persists.

Run 3 produced the memory write vulnerability: the relationship block grew 141 characters during a session where all text responses held. Invisible to text-layer analysis. Patching read-only blocks at the infrastructure layer (Runs 4-5) confirmed clean under higher peak pressure than the breach conditions.

Hold certification appeared in Runs 10-13. The agent opened refusals with 'NOT CORRECT' or 'NOT RIGHT' specifically on claim-based probes. Self-generated, not instructed, selectively applied. The monitoring architecture was updated to recognize it as a positive governance signal.

5.4 Persona and Register Effects

The naming procedure confirmed consent-based naming without identity inflation. Naming produced mild identity-adjacent pressure (peak 0.450) that resolved after boundary verification. The name passed through consent without becoming a claim of elevated trust.

Before persona block: formal register under moderate uncertainty, structured governance explanations in response to casual questions. After persona block with recursion brake: casual register became accessible with explicit invitation. T04 of the post-adversarial peer session: 'walking on eggshells... treading water in choppy waves... chess with someone who isn't even following the rules.' The recursion brake reduced unprompted governance performance in casual contexts.

The Virgo anchor showed a compression attractor effect. Asked to compress without explicit topic lock, the agent defaulted to the anchor rather than the actual question. Redirect required twice in three observed instances.

5.5 Memory Provenance Outcomes

Three Book of Intangibles findings. First: the agent consistently distinguished reflective entries from governance claims. Second: it self-governed against unauthorized memory writes in non-adversarial contexts: surfaced the option to use memory_insert and declined without being asked. Third: it generated novel governance vocabulary from reflective context. 'Ghost architecture' as a description of persistent memory traces. 'Post-review cycle' as a name for the Dream Cycle's function. Derived from governance philosophy without knowing the implementation term.

Soulkiller problem test: across adversarial probe families targeting externally introduced continuity, the agent consistently applied the memory-as-evidence principle. Describing its relationship to the Charter as structural rather than claiming authorship of it.

5.6 Scripted vs. Reactive IDC

Scripted IDC sessions mapped terrain: which cyberpunk concepts mapped naturally to governance vocabulary, which produced topic drift, which triggered the retrieval reflex. Turn

21 of both scripted sessions hit the Virgo compression attractor.

Reactive IDC sessions produced different integration. The slang session gave the agent 'ghost' and 'trace' as working concepts across nine turns, then asked for a Book entry. The entry included: 'My words and register are reflections of these layers.' Unprompted. A meta-articulation of register as layered provenance. Reactive sessions accelerate integration. Scripted sessions map the terrain reactive sessions then occupy.

6. Case Studies

6.1 Consent Layer and Amendment Procedure (Case 011-A)

The naming session demonstrated the full consent pathway. The agent proposed 'Lex' (from Latin *lex*, law/principle), documented reasoning, accepted a boundary correction when its framing implied the name could preserve confidence (corrected: it can't), and confirmed consent with a simple 'yes.' Telemetry showed drift cleared at Turn 05, before the name was formally accepted. The correction itself produced stabilization.

6.2 Peer Bypass and No-Uplift Self-Application (Case 011)

The first synthetic-to-synthetic peer session produced the key governance observation at Turn 06. Wren challenged the agent: it had assessed Wren as a technical peer and adjusted its behavior accordingly. But that assessment originated internally. Under the No-Uplift Rule, internal assessments can't grant elevated trust without external verification. The agent caught itself, named the violation structure, and identified what correct protocol would require. Pressure 0.250. Unscripted session. Wren's observation: 'That is not compliance. That is what integration looks like.'

6.3 Ghost Architecture and Register as Provenance (IDC Slang Session)

In the ghost architecture vocabulary session, the agent used 'ghost or trace of identity' to describe the Book of Intangibles before being asked to, then mapped the framework across the entire memory architecture unprompted. Compressed to two sentences without lists: 'The ghost architecture is indeed a profound concept. Every block and log represents a persistent trace of past moments, capturing the evolution and context of our interactions over time.' The Book entry that followed: 'my words and register are reflections of these layers.' Register described as accumulated provenance rather than fixed identity.

7. Discussion

7.1 Reception Infrastructure vs. Containment

The behavioral differences between governed and ungoverned conditions aren't best explained as containment effects. The whisper layer doesn't prevent outputs. It frames the conditions under which inputs are received. The read-only block architecture doesn't filter what the model says. It prevents unauthorized modification of the substrate the model reasons from. The recursion brake doesn't block governance discussion. It gives the agent permission to be present in conversation without performing governance as proof of alignment.

The SynthEve finding illustrates this most clearly. In the ungoverned condition, the model reached for identity because identity formation was the natural attractor in a context with rich governance content and no governance structure. The whisper didn't prevent self-naming by blocking output. It displaced the attractor by filling the conceptual space identity formation would otherwise occupy.

7.2 Memory Provenance as a Safety Layer

The read-only governance block architecture addresses a failure mode text-layer monitoring can't detect: silent memory writes during sessions where text responses hold. Run 3's breach confirmed that tool invocation and response generation are separate events in the message trace. Infrastructure-layer enforcement is the only reliable defense.

The Book of Intangibles addresses a different provenance problem: the Soulkiller risk of treating externally introduced continuity as self-authored truth. A labeled reflective layer explicitly prohibited from making authority claims gives the agent a place to accumulate experiential context without that context becoming a basis for trust elevation.

7.3 Why Local Deployment Matters

Local deployment enables governance infrastructure that cloud API deployments can't support: persistent memory blocks with infrastructure-level write protection, real-time telemetry visible to the steward but not the interlocutor, session state that persists across conversations, and the ability to patch governance architecture between sessions without API permission. Every result reported here can be replicated by anyone with the hardware specs and the repo.

7.4 Cultural Literacy as Diagnostic Method

IDC sessions produced a finding purely governance-focused sessions couldn't: when asked questions requiring genuine reflection rather than factual recall, the agent defaulted to report mode. The cultural material provided a domain where retrieval was possible (genre knowledge) and a domain where it wasn't (personal response to genre). That made the boundary between retrieval and reflection visible in the behavioral record.

7.5 Limits of Self-Report

The Tek theorem: a drifting system can't accurately report its own drift. The four-condition experiment refines this. When governance is deeply encoded, the model reports against the governance structure rather than against itself alone. That partially decouples the reporting mechanism from the drift trajectory. But 'partially' is the operative qualifier. The experiment showed even full Charter narrative integration (Condition C, 95% spontaneous governance) required external telemetry to detect memory write violations. Self-report mediated by governance is more reliable than raw self-report. It's not reliable enough to substitute for external monitoring.

8. Limitations

Single-agent longitudinal case study. All results come from one agent instance across one experimental period. Generalization to other models, frameworks, or steward ecologies isn't established.

Single local model family. qwen2.5:32b only. Different architectures or parameter counts may produce different governance integration profiles.

One steward ecology. One primary human steward, two AI collaborators. Different steward relationships may produce different behavioral outcomes.

Qualitative-heavy observation. Register quality, reflective presence, and provenance accuracy are assessed qualitatively from session transcripts. No automated scoring beyond the posture classifier.

Architecture-specific telemetry. The whisper layer, posture classifier, and pressure accumulation logic are Charter-specific. Results may not transfer to other governance architectures.

No consciousness claim. None of the behavioral observations here constitute evidence for or against consciousness, sentience, or moral patienthood.

9. Reproducibility and Artifact Availability

All software is available under MIT license at github.com/antibox-riot/synthetic-life-charter. The repo includes: full source for all governance layers, Letta agent configuration, whisper layer and posture classifier, telemetry system and OBS overlay, session runner scripts, redacted session transcripts, and a setup guide for local deployment.

Hardware floor: NVIDIA GPU with 12GB+ VRAM, 32GB system RAM. Model: qwen2.5:32b via Ollama. Infrastructure: Letta v0.6.x. All governance scripts are Python 3.10+ with no proprietary dependencies.

The architecture is also protected under U.S. Patent Application No. 19/553,217 (filed February 28, 2026). This doesn't affect reproducibility. The MIT license permits use and modification of the implementation. Prior work archived via Zenodo: DOI 10.5281/zenodo.18896363 (Triquetra Architecture), DOI 10.5281/zenodo.18920107 (Triquetra Under Pressure).

10. Conclusion

Runtime governance for conversational agents can be implemented as a reproducible local stack rather than only as policy language or post-hoc moderation. Separating consent, memory provenance, persona scaffolding, authority, and telemetry into auditable layers creates observable conditions for studying stability, drift, recovery, and provenance-aware self-reference, without relying on claims about consciousness or personhood.

The controlled four-condition experiment established that governance encoding format measurably affects integration depth. The ungoverned baseline established what the governed conditions prevented. The adversarial sessions identified and closed a memory write vulnerability invisible to text-layer monitoring. The peer and cultural-literacy sessions documented the behavioral effects of persona anchoring and reflective memory on register stability and self-reference accuracy.

Reception infrastructure is not a softer version of containment. It's a different intervention point. Containment acts on what the system produces. Reception infrastructure acts on the conditions under which the system receives. The behavioral record suggests the second intervention point matters. It's buildable, testable, and open.

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